Towards real-time calibration-free LIBS supported by machine learning

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Abstract

Calibration-Free Laser-Induced Breakdown Spectroscopy (CF-LIBS) enables multi-elemental quantification without needing standards. This type of approach can be used to analyze complex samples containing traces or gradients of species. This type of diagnosis requires a high level of expertise, and is cumbersome to set up. These constraints limit its application to field diagnostics. Using the MERLIN generalized radiative transfer code, we are able to generate a diversified emission database with no dimensioning limitations. We show that training a convolutional residual network with such a database enables the quantification of 9 species, as well as evaluation of electron density and temperature, without any prior expertise at a rate greater than 10 Hz. The accuracy of this innovative method depends solely on the basic spectroscopic data (emission probabilities and Stark parameters), regardless of the thermodynamic conditions of the laser-induced plasma, as long as it is in Local Thermodynamic Equilibrium (LTE).

Keywords: LIBS, CF-LIBS, Real-time, Quantification, Radiative transfer, Machine Learning, ML, Artificial Intelligence, AI

1. Introduction

Spectroscopic analysis of laser-induced plasma emission is known as LIBS (laser-induced breakdown spectroscopy). The recombination dynamics of this type of plasma are strong. Its electron density n_e and temperature T are respectively of the order of 10^{27} m^{-3} and 10^5 K during the first nanoseconds of expansion, ending up at 10^{21} m^{-3} and 10^3 K at the end of the dynamics ($\sim$ 10 µs). During these dynamics, there is a time window in which plasma thermalization is assured: this is known as local thermodynamic equilibrium (LTE). From the spectral emission of this time window of a few µs, it is possible to analytically associate the population densities of the emitting levels with the total species density of each plasma element [1]. Observations made in this time range can then be exploited through 4 different strategies :

1. qualitatively: identification and speciation,
2. semi-quantitatively: estimation of species ratios,
3. quantitatively with calibration: comparison with emission from representative samples,
4. quantitatively without calibration: simulation of calibrated emission in spectral radiance L_λ (W m^{-2} sr^{-1} m^{-1}), known as *Calibration Free*-LIBS (CF-LIBS).

Machine learning approaches are used to approximate a problem's solution using experimental or simulated data. In this context, strategies N° 1, 2 and 3 of LIBS signal exploitation can be considered as classification problems aiming at identifying a species in a spectrum or associating a family of spectra to a material [2][3]. Strategy N°4, on the other hand, is similar to a regression problem to associate a signal level with one or more target quantities. For multi-element quantification, it is about predicting the $\mathbf{x_i}$ vector of mole fractions from a $\mathbf{L}_\lambda$ vector whose spectral components are strongly non-linearly associated with the values of n_e, T and the x_i components. Based on experimental non calibrated signal level, pioneering works related to the (n_e, T) diagnosis [4] and the prediction of growth curves [5][6][7] have to be cited. To our knowledge, the only published work on this subject is based on indicative emission without considering emitting plasma geometry (no radiative transfer) or physical broadenings (Stark predominant for laser-induced plasmas, Doppler sometimes non-negligible depending on the element considered) [8]. Although promising, the representativeness of this type of dataset does not allow us to be predictive on field signals.

Our team has developed a generalized radiative transfer code called MERLIN (MultiElemental Radiative equiLibrIum emissioN) [9]. This code can be used to simulate the emission of any mixture under any LTE conditions [10][11]. In this paper, we propose a preliminary study of deep learning based on a set of synthetic spectra generated using the MERLIN code. First, we briefly present the characteristics of the database, the network and the training and validation modalities. We then show that this training leads to a satisfactory sensitivity for multi-element quantification. Finally, we discuss the multi-parametric dependence of this network sensitivity, showing that current optimization already enables real-time quantification.

2. Methodology

The reader is invited to refer to tables in appendices for the notations used in the following.

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2.1. Dataset

The MERLIN code enables the emission simulation of inhomogeneous plasmas whose different layers can have any LTE composition under any thermodynamic conditions. This type of code allows to generate a realistic emission database much easier and faster than experimentally [10][11].

The simplest physical situation corresponds to a homogeneous plasma. The expression of its spectral radiance is obtained by integrating the 1D radiative transfer equation at steady state and without diffusion:

$$L_\lambda(\mathbf{x_i}, n_e, T) = L^0_{\lambda,T}\left(1 - e^{-\alpha_\lambda(\mathbf{x_i}, n_e, T)\,\cdot\,\ell_c}\right) \tag{1}$$

where $L^0_{\lambda,T}$ (W m^{-2} sr^{-1} m^{-1}) is the spectral radiance of the blackbody at temperature T (K) and ℓ_c (m) is the core dimension of the emitting plasma. In this expression, α_λ (m^{-1}) is the spectral absorption coefficient dependent on the mole fractions x_i of each species present in the plasma (LTE composition at given T and electron density n_e) and their respective radiative properties. This coefficient results from the sum of many contributions, each being characterized by a hybrid Voigt profile mainly due to Stark, Doppler and the apparatus broadenings.

For the test case presented in this paper, we consider a homogeneous plasma of dimension ℓ_c = 750 µm singly ionized (X I and X II) and a Lorentzian apparatus function with a full width at half maximum set to 10^{-1} nm representative of the experimental configuration. Spectroscopic data are taken from the NIST database [12]. With the exception of hydrogen (empirical Gigosos law [13][14]), the contribution to Stark broadening is estimated using a polynomial correlation [15]. The dimensioning of this dataset is conditioned by the variety required for efficient learning. This variety is driven by the following three parameters.

- The $\mathbf{x_i}$ vector corresponds to the mole fractions of 9 distinct elements (Ag, Ar, B, C, Cu, H, Mo, Ni and W). These elements have been selected in accordance with analyses carried out within the WEST tokamak [16]. We have added Ar, whose confinement properties on the laser-induced plasma makes it suitable for LIBS diagnostics, and it is therefore being considered for future *in-situ* measurements [10]. By definition, these random mole fractions have a sum egal to 1. Their generation is therefore performed in a Dirichlet space [17]. We set the minimum value of the $\mathbf{x_i}$ components to 10^{-6} (1 ppm). In this way, we obtain the lognormal distributions shown in Fig. 1. The distributions of x_{Ni} and x_H are shown in red and blue respectively. The identification of these two particular species is discussed in section 3.2. The mean distribution over the 9 species, shown in black, correlates very well with those of Ni and H. We note that, among the vectors generated, it is very unlikely to obtain minor species ($x_i \lesssim 10^{-3}$) or major species ($5 \times 10^{-1} \lesssim x_i$). Indeed, the development of this approach has shown that effective learning requires that these configurations do not have a probability of zero, and that the data set has to be as varied as possible. Finally, we note that the average mole fraction is equal to 1/9 (vertical black dotted line).

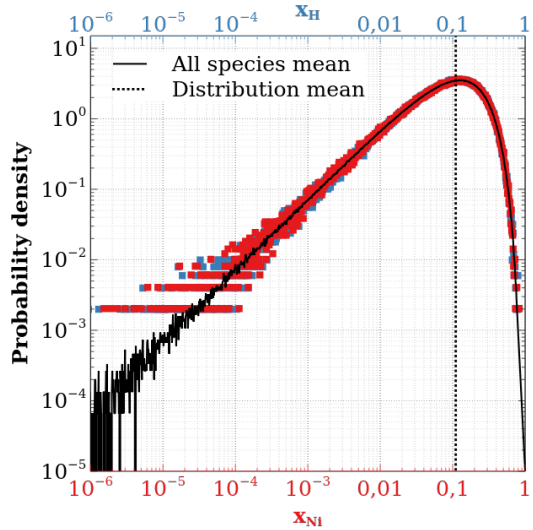

Figure 1: Mole fractions $\mathbf{x_i}$ probability densities (30% random sample of the whole dataset - each point results of the integration of 1000 spectra). x_{Ni} and x_H are respectively representated in red and blue. The all species mean distribution is representated in black. The mean value of x_i is shown in black dotted line.

- Training is performed over the [200, 800] nm spectral range. For each spectrum in the dataset, we randomly draw the central wavelength λ_c from a uniform distribution whose extrema are 230 nm and 770 nm. We fix the width of the spectral interval at 60 nm with a discretization of the order of 5×10^{-2} nm (low-resolution broadband experimental setup). The λ_c probability density function obtained is shown in green on Fig. 2. We note a mean value of ≈ 500 nm (center of the domain [200, 800] nm).

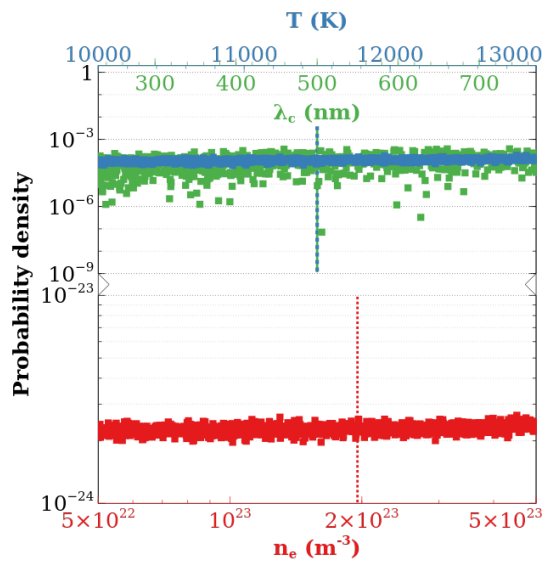

Figure 2: Center wavelength λ_c (nm, in green), electron density n_e (m^{-3}, in red) and temperature T (K, in blue) probability densities (30% random sample of the whole dataset - each point results of the integration of 1000 spectra). All of those three parameters distribution are uniform. The dotted lines stand for mean values of each variable.

- The thermodynamic conditions for plasma emission are given by n_e and T. For these two quantities, we respectively consider the intervals [5, 50]×10^{22} m^{-3} (logarithmic space) and [1, 1.3]×10^4 K (linear space). The extrema of these intervals correspond to current values measured in argon or on metal matrices at laser irradiances suitable for LIBS diagnostics [14][18]. The probability density functions obtained are respectively shown in red and blue on Fig. 2. The associated mean values (red and blue dotted lines) correspond to the centers of the intervals considered. The physical coupling between n_e and T is imposed in accordance with Fig. 3. The choice of this coupling is arbitrary in that, to our knowledge, no paper reports plasma parameter measurements for a mixture of these 9 species. Nevertheless, the power law interpolation (in orange) is compatible with previous measurements [19][20].

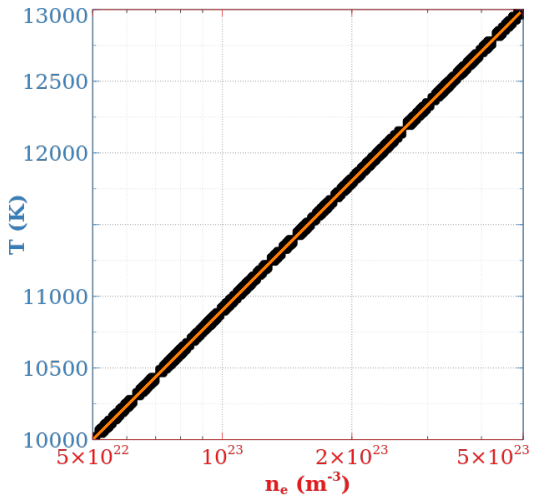

Figure 3: Temperature T (K, blue axis) versus electron density n_e (m^{-3}, red axis). A linear correlation between T and n_e is arbitrary imposed.

The database Ω was built from 10^6 vectors storing wavelengths (Ω_{λ_c}), spectral radiances (Ω_{L_λ}) and target values of $\mathbf{x_i}$ (Ω_{x_i}), n_e (Ω_{n_e}), T (Ω_T) and λ_c (Ω_{λ_c}). The computation of Ω was performed in ~ 1 week on several dozen parallel CPUs. Before training the network, each part of Ω must be normalized in different ways. For inputs corresponding to radiance data :

1. Before normalization, L_λ values within Ω_{L_λ} range from ~ 5×10^9 to ~ 10^{15} W m^{-2} sr^{-1} m^{-1}. To avoid scaling effects, we calculate $\Omega_L = \log(\Omega_{L_\lambda})$.
2. The statistical study of Ω_L allows us to extract the arithmetic mean $\bar{\Omega}_L$ and the standard deviation σ_{Ω_L}.
3. Finally, the logarithmic dataset is normalized according to the relationship $\Omega_L^* = (\Omega_L - \bar{\Omega}_L)/\sigma_{\Omega_L}$ so that $\bar{\Omega}_L^* = 0$ and $\sigma_{\Omega_L^*} = 1$ (Z-score). The values within Ω_L^* then range from -1.344×10^{-1} to 4.672×10^1.

Inputs of $\mathbf{x_i}$, n_e, T and λ_c are proceeded as follows.

1. Mole fraction are redistributed into a log ratio space (Eq. 2) allowing a more efficient generalization.

$$\mathbf{x}_i^* = \log(\mathbf{x}_i) - \frac{1}{9}\sum_{j=1}^{j=9} \log(\mathbf{x}_j) \qquad (2)$$

The values within $\Omega_{x_i^*}^*$ then range from −9.719 to 3.277.

2. Z-score normalization of n_e : the values within $\Omega_{n_e}^*$ then range from −1.163 to 2.467.
3. Z-score normalization of T : the values within Ω_T^* then range from −1.725 to 1.744.
4. Z-score normalization of λ_c : the values within $\Omega_{\lambda_c}^*$ then range from −1.719 to 1.741. This output is only use for us to develop the neural network.

The data used for training are extracted from Ω^* which is the concatenation of normalized (exponent $*$) data.

2.2. Network

An emission spectrum is a set of consecutive ordered radiances each related to specific wavelengths. Thus, emission spectra are structured data. Multi-Layers Perceptron (MLP) do not preserve data structure contrary to Convolutional Neural Network (CNN). For this reason, we developped a CNN specifically tuned for the analysis of LIBS spectra. The input is a single vector of size 10240 containing the radiance from Ω_L^* over the whole [200, 800] nm spectral range filled with zeroes outside the 60 nm window around λ_c. The neural network architecture adopted is divided into two parts. The first part, composed of 3 stages, is devoted to the extraction of relevant features. Each stage allows to extract important features by reducing the dimension of the data and increasing its number of channels (for a given input spectrum each channel highlights different information) with a simple convolution operation and a residual module. The second part, built of 3 dense layers, is devoted to the regression. The nodes of the whole structure result in ~ 6×10^7 weights and bias. The output is composed of 3 vectors respectively containing : the 9 species mole fractions, the 3 physical properties (n_e, T and λ_c) and the 12 confidences associated with the $i_{max} = 9 + 3$ predicted variables.

2.3. Training and test

The network was trained on random 10% of Ω^*. This sample has been split in 90% (9×10^4 spectra) for training and 10% (10^4 spectra) for testing. The learning process has been operated with batchs of 32 normalized spectra. This value was chosen according to 1) moderate computing time and 2) memory capacity of the used GPU (the GPU needs to store the model, the spectra, and the computation of the gradients). The learning rate was set to 5×10^{-5} and the number of epochs is fixed to 100. We used a customized confidence-based mean square error loss function :

$$\mathcal{L}(\mathbf{p}, \mathbf{t}, \mathbf{c}) = \sum_{i}^{i_{max}} c_i^2 (p_i - t_i)^2 + \sum_{i}^{i_{max}} (1 - c_i)^2 \qquad (3)$$

where i is the variable of interest (for i_{max} explanations, see section 2.2) p_i is the prediction of i, t_i is the target value of i and c_i is the prediction confidence. From an analytical point of view, we note that $\mathcal{L}$ is minimized if $(p_i - t_i)$ is minimized and if c_i tends to 1. Figure 4 shows the evolution of $\mathcal{L}$ with respect to the number of epochs. The error is decreasing over epochs both for train data (red points) and test data (blue points). It does

not seems to exhibit over-fitting behavior, suggesting that the network is still learning while remaining able to generalize over unknown spectra after 100 epochs. Training for more epochs might give us slightly better results.

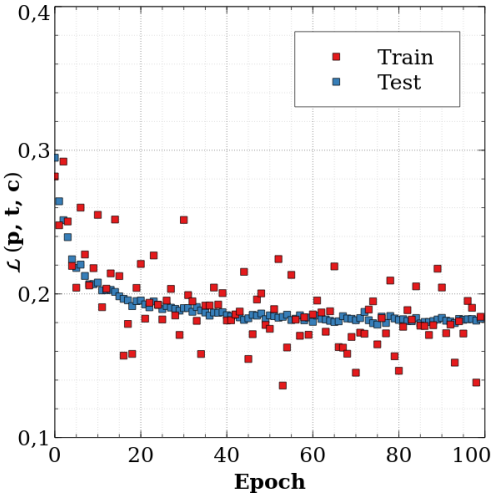

Figure 4: Custom loss $\mathcal{L}$ versus to the number of epochs. Red dots corresponds to the error on the last batch of train data while blue dots corresponds to the error on all test data.

We are currently exploring the best architecture settings, the best set of hyper-parameters, the optimal size of the database to find an optimized training configuration for our network. Due to the time to train (on one commercial GPU), we cannot fully train our network with the best performance. In the future, we will use a super-calculator with multiple nodes and multiple GPUs to train our network. Nevertheless, we show in the next sections that our trained network is already able to provide good results.

3. Preliminary results

The results presented in this section are preliminary. We here evaluate the network's ability to interpret and analyze synthetic signals. They are obtained from 10^3 spectra (0.1% of Ω^*) randomly selected from the 90% of Ω^* not used for training or validation. This reduced sample is referred as $\breve{\Omega}^*$ in the following. We define the following three metrics in support of discussions of our results.

- $c_P(i)$: Pearson correlation coefficient between predictions and expected values of variable i.
- $\bar{e}_r(i) = \frac{1}{N_k}\sum_k \left(\frac{|i_p - i_t|}{i_t}\right)_k$: mean absolute deviation between N_k predictions p and target values t of variable i.
- $\bar{c}(i) = \frac{1}{N_k}\sum_k (c_i)_k$: mean confidence in the N_k predictions of variable i.

3.1. Plasma diagnostic

At present, (n_e, T) estimation is essential for CF-LIBS. The measurement of n_e is operated by analyzing the Stark contribution to the spectral profile of given transitions. To obtain an accurate value of the temperature T, it is necessary to establish a diagram (Boltzmann or Saha-Boltzmann) of the total radiances of several transitions of the same element. In addition to spectroscopic expertise (baseline estimation, self-absorption, choice of transitions of interest), determining the pair (n_e, T) therefore requires deconvolution and integration operations.

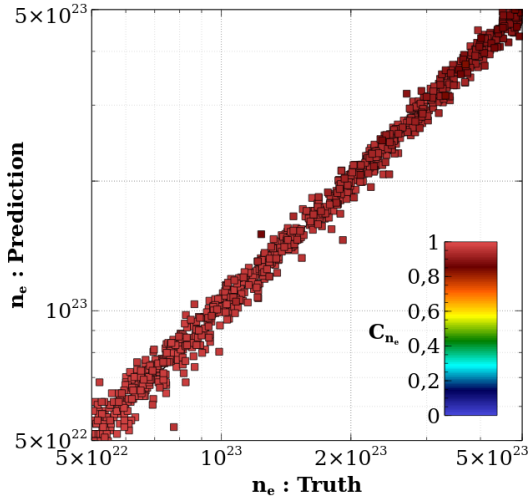

Figure 5: Predicted n_e versus expected n_e for $\breve{\Omega}^*$ spectra. The colormap represents the n_e confidence.

In order to test the robustness of our training, we defined n_e and T as target output values of the network. We show in Fig. 5 and Fig. 6 respectively, the predictions of n_e and T as a function of the expected values. For the sample $\breve{\Omega}^*$, we obtain respectively for n_e and T, $c_P(n_e) = 0.99628$ and $c_P(T) = 0.99627$ (linear trend between predicted and expected values). On the other hand, $\bar{e}_r(n_e, T) < 0.05$ (0.04546 for n_e and 0.00516 for T) and $\bar{c}(n_e, T) > 0.95$ for these two variables. This results in a short distance to the first bisector and uniform coloring of the points in Fig. 5 and Fig. 6.

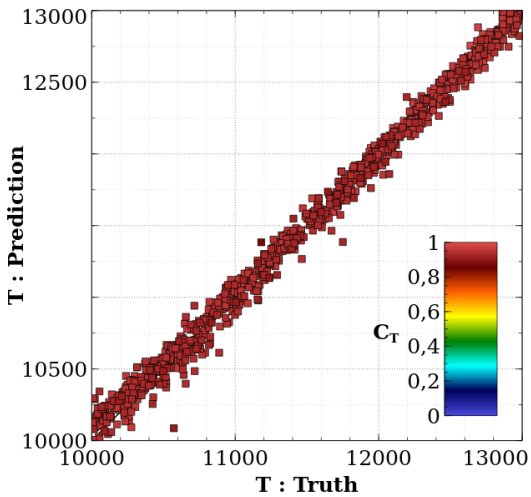

Figure 6: Predicted T versus expected T for $\breve{\Omega}^*$ spectra. The colormap represents the T confidence.

Thus, our network is able to determine (n_e, T), regardless of the components $x_i \in [10^{-6}, 1]$ of $\mathbf{x_i}$, whatever $\lambda_c \in [230, 770]$ nm, whatever the values of $n_e \in [5, 50] \times 10^{22}$ m^{-3}

and $T \in [1,\ 1.3] \times 10^4$ K with an average accuracy of less than 5%.

3.2. Elemental quantification

For the sake of simplicity and pedagogy, we cannot develop the network's response to all 9 species. We have therefore chosen 2 species that are representative of the analytical capabilities of this training. The choice of these two species is based primarily on their electronic structure, and therefore on the emission wavelengths of their transitions. Indeed, for nickel, NIST reports 493 Ni I transitions and 143 Ni II transitions that can be simulated on [200, 800] nm [12]. These transitions are uniformly distributed over this interval. In addition, the Ni ionization potential (7.6398 eV) is favorable to the generation of Ni^+ and thus to the emergence of Ni II transitions at low (n_e, T) values. Conversely, there are few intense hydrogen transitions (Balmer series) and they are spectrally localized (notably H_δ 410.17 nm, H_γ 434.08 nm, H_β 486.13 nm and H_α 656.28 nm). In addition, H^+ does not radiate. The ionization of H tends to reduce the radiative signature of H at a fixed x_H.

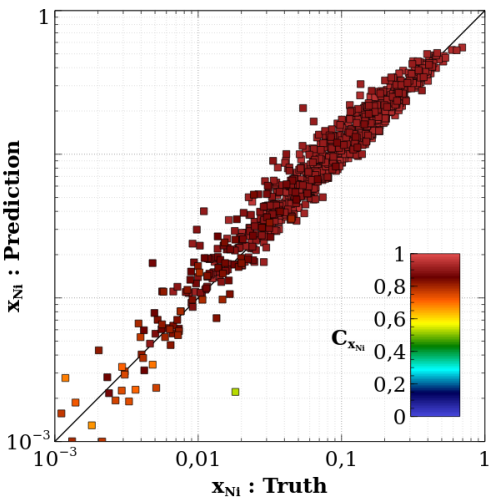

Figure 7: Predicted x_{Ni} versus expected x_{Ni} for $\breve{\Omega}^*$ spectra. The colormap represents the x_{Ni} confidence.

Figs. 7 and 8 show how sensitive the network is to this electronic structure difference. For Ni (Fig. 7), the distribution is monomodal with metrics $(c_P, \bar{e}_r, \bar{c})(x_{Ni}) = (0.96535, 0.26744, 0.91626)$ whereas, for H (Fig. 8), the distribution is bimodal. This is reflected in the metrics $(c_P, \bar{e}_r, \bar{c})(x_H) = (0.56307, 3.36193, 0.58012)$. Points of low confidence are far from the first bisector: confidence c_{x_H} is therefore a good indicator for both the network and the analyst.

The representation of the unaveraged absolute deviation $e_r(x_H) = \frac{|x_{H_p} - x_{H_t}|}{x_{H_t}}$ as a function of λ_c on the $\breve{\Omega}^*$ dataset (Fig. 9) particularly highlights the spectral sensitivity of the network to hydrogen analysis. Indeed, we find the bimodal distribution of low and high confidence points. High-confidence points correspond to spectral domains where at least one of the transitions in the Balmer series is observable. The horizontal green

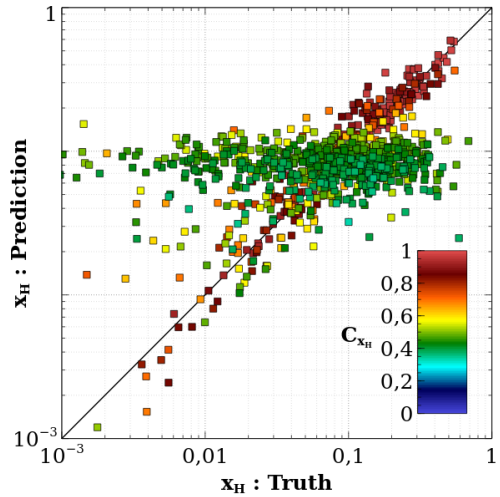

Figure 8: Predicted x_H versus expected x_H for $\breve{\Omega}^*$ spectra. The colormap represents the x_H confidence.

dotted line corresponds to $\bar{e}_r(x_H) = 3.36193$. The dotted red horizontal line corresponds to the mean relative deviation of spectra for which $630 < \lambda_c(\text{nm}) < 670$: the associated value is 0.11174. We can therefore see that spectrally filtering the spectra to guarantee the presence of H_α reduces the deviation from the expected value by a factor of ~ 30.

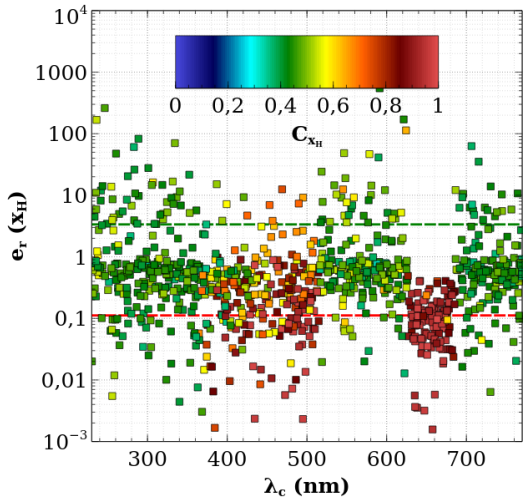

Figure 9: Relative error $e_r(x_H)$ versus center wavelength λ_c computed from predicted x_H and expected x_H for $\breve{\Omega}^*$ spectra. The colormap represents the x_H confidence.

So we define a new sample $\breve{\Omega}^*_f$ obtained by randomly drawing 10^3 spectra of which $630 < \lambda_c(\text{nm}) < 670$ from the 90% of Ω^* that were neither used for learning nor validation. In this filtered sample, the results for Ni are similar: $(c_P, \bar{e}_r, \bar{c})(x_{Ni}) = (0.98531, 0.14044, 0.93142)$. The slight narrowing of the distribution around the first bisector (better Pearson coefficient and lower mean error than the results of Fig. 7 on $\breve{\Omega}^*$) is synonymous with a gain in sensitivity for Ni quantification at these wavelengths. With regard to H, we show in Fig. 10 the predictions obtained on the $\breve{\Omega}^*_f$ sample. This time

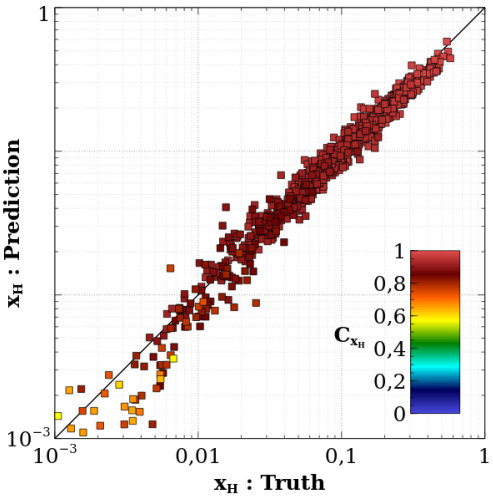

Figure 10: Predicted x_H versus expected x_H for $\breve{\Omega}^*_f$ spectra. The colormap represents the x_H confidence.

we see a monomodal distribution with a clear improvement in all three metrics: $(c_P, \bar{e}_r, \bar{c})(x_H) = (0.98425, 0.18144, 0.92936)$. For the 10^3 spectra of $\breve{\Omega}^*_f$, we find a slightly higher average error (diversity of pairs (n_e, T)) but close to that calculated from the 83 spectra of $\breve{\Omega}^*$ (average in red dotted lines in Fig. 9). Finally, from a quantitative point of view, the lowest confidence is found for $x_H \lesssim 10^{-2}$. All this shows that the network can only perform reliable quantification in the presence of an explicit radiative signature of the probed element.

4. Discussion about the performance

Given the size of $\breve{\Omega}^*$ and the several parameters involved, we discuss as exhaustively as possible in this section their impact on the multi-elemental quantification. As mentionned in section 2, the network is asked to predict λ_c for development purpose. Since this prediction is not relevant for this paper topic, we only give the metrics obtained on $\breve{\Omega}^*$ showing the reliability of the λ_c prediction with no additional discussion : $(c_P, \bar{e}_r, \bar{c})(\lambda_c) = (0.99933, 0.01076, 0.97572)$.

4.1. Spectral range

The cases of Ni and H developed in section 3.2 are representative of the spectral response of the network. Indeed, among the 7 other species, some, like Ni, have a uniform distribution of e_r with λ_c and others, like H, have spectral regions more favorable to quantification. The species that stand out as analogous to Ni are Mo and W. Their electronic structure is similar to the one of nickel (alike ionization potentials and numerous electronic levels leading to emission uniformly distributed over $[200, 800]$ nm). As for the other 6 species, their quantification is highly dependent on the spectral range considered. As can be seen in Table 1, for each of these species there is a strong correlation between these preferred spectral intervals and the most intense transitions listed by NIST [12].

The statistical study of all the predictions made on $\breve{\Omega}^*$ shows that the sum of the components of $\mathbf{x_i}$ at the network output is, on average, very close to the unit (to the nearest $\approx 10^{-15}$, machine precision), without imposing this condition beforehand. Let us consider the average algebraic deviation between the expected and predicted x_i ($\frac{1}{9}\Sigma_i \frac{x_{ip}-x_{it}}{x_{it}}$). From this metric, we can select the ten best predictions of the network on the simultaneous quantification of the 9 species (Fig. 11). We therefore explain, by complementarity, that among these ten best predictions made on the unfiltered set $\breve{\Omega}^*$, eight of them are made on a spectrum containing the emission of H_α.

We detail the spectrum No 405 arbitrarily chosen among these ten predictions. The input values as well as the network output predictions are given in Table 2. The average algebraic deviation between the expected and predicted x_i is $\approx -0.1\%$. This small difference must be put into perspective to the extent that, always by complementarity, the mole fractions of certain underestimated species can be compensated by certain other overestimated ones (and vice versa). As such, the absolute average deviation (here $\approx 6.5\%$) can also be a metric of interest.

Element	λ_c range(s) (nm)	NIST strong line(s) (nm)
Ag	[230, 350]	Ag II 246.64
		Ag II 241.32
		Ag II 243.78
	[450, 500]	Ag II 478.84
Ar	[660, 770]	Ar I 696.54
		Ar I 706.72
		Ar I 750.39
		Ar I 763.51
		Ar I 794.82
B	[230, 280]	B I 249.77
	[580, 650]	B II 608.04
C	[230, 280]	C I 247.86
	[600, 770]	C II 657.80
		C II 723.64
Cu	[300, 350]	Cu I 324.75
		Cu I 327.39
	[600, 770]	Cu II 627.33
		Cu II 740.43
		Cu II 780.76

Table 1: Comparison between low $e_r(x_i)$ / high $\tilde{c_{x_i}}$ spectral ranges and main NIST most intense lines.
The best network analytical power is well correlated to NIST strong lines.

Variable	Truth	Prediction	Unit	$\frac{\text{Prediction}-\text{Truth}}{\text{Truth}}$ (%)
n_e	2.506	2.495	10^{23} m^{-3}	-0.41
T	12.099	12.092	10^3 K	-0.58
x_{Ag}	0.123	0.141		14.43
x_{Ar}	0.020	0.017		-16.81
x_B	0.099	0.106		7.53
x_C	0.115	0.111		-3.87
x_{Cu}	0.185	0.180		-2.60
x_H	0.142	0.134		-5.11
x_{Mo}	0.222	0.218		-1.74
x_{Ni}	0.029	0.030		3.07
x_W	0.065	0.063		-3.43

Table 2: Truths and predictions for the spectrum No 405 from Fig. 11.

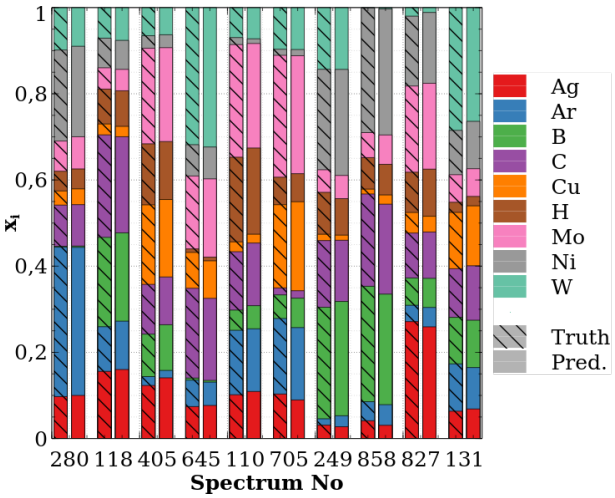

Figure 11: Comparison between truth and predictions for the ten best (based on lowest x_i algebrical mean error) quantifications operated on $\check{\Omega}^*$. The spectrum number is between 1 and 1000 ($\check{\Omega}^*$ dimension).

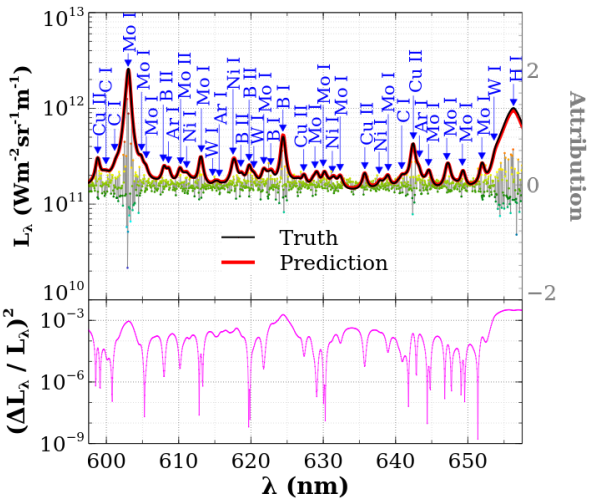

Figure 12: Truth (in black) and predicted (in red) spectra obtained from Table 2 values. Line identification (in blue) is operated only for significant lines (total of 302 NIST lines on [597.55, 657.51] nm spectral range). Network spectral attribution (in grey and colored points) is computed from integrated gradient method. The quadratic deviation is exposed in purple.

We show in Fig. 12 the comparison between the input spectrum (No 405 in black, "Truth" column of Table 2) and the spectrum associated with the network predictions (in red, "Prediction" column). The quadratic relative deviation in the lower part of the figure shows the very good agreement between the two spectra. The largest gaps appear around 624.5 nm (overestimation of x_B, emission of B I 624.46 nm triplet) and in the form of a plateau around 656.3 nm (significant Stark broadening of H_α and slight underestimation of x_H by the network). Furthermore, the spectral interval [597.55, 657.51] nm is characterized by a radiative signature where most of the species are represented (see Table 1). The highest differences observed in Table 2 are associated with Ag (absence of emitting transition) and Ar (few intense transitions and low quantity). Conversely, Mo, Ni and W are quantified with a better precision because of their large number of transitions over this interval (respectively 45, 25 and 33 transitions, neutrals and ions combined). Finally, we represent in gray the sensitivity of the network at each wavelength for the No 405 spectrum. This metric, obtained using an *integrated gradient* approach, accounts for the wavelengths most likely to affect the response of the network, and therefore the quantification [21]. Generally speaking, in addition to the level of L_λ, the network appears sensitive to $\frac{dL_\lambda}{d\lambda}$. This higher order sensitivity is explained by the convolutional layers which here make it possible to extract information in the case of spectral overlapping or low spectral radiance values.

4.2. Thermodynamical parameters

If sections 3.2 and 4.1 show that the reliability of the predictions is very strongly correlated to λ_c, the response of the network to the values of n_e and T is quite different. We show in Fig. 13 the unaveraged absolute deviation $e_r(x_{Ni})$ as a function of n_e. The choice of the variable x_{Ni} is motivated by the weak spectral dependence of the nickel quantification. This makes it possible to avoid analysis biases, in particular the interference of the role of λ_c with the couple (n_e, T) on the prediction. We see a uniform distribution of $e_r(x_{Ni})$ according to n_e. The horizontal orange dotted line corresponds to $\bar{e_r}(x_{Ni}) = 0.19573$. For all of these points, the confidence is high: 0.91626 ± 0.04814. Also, this figure highlights that the quality of the predictions does not depend on the value of the electron density.

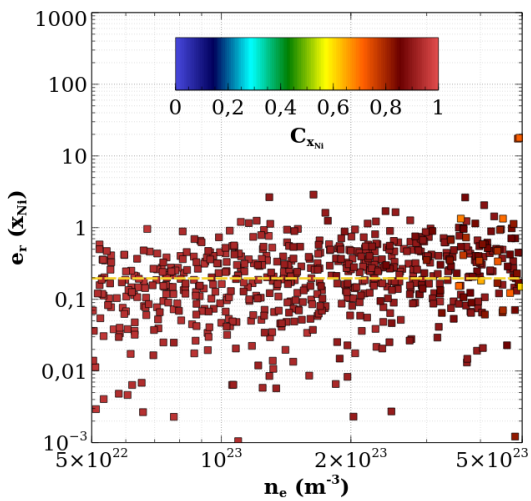

Figure 13: Relative error $e_r(x_{Ni})$ versus center wavelength n_e computed from predicted x_{Ni} and expected x_{Ni} for $\check{\Omega}^*$ spectra. The colormap represents the x_{Ni} confidence.

The Ω^* dataset only contains LTE spectra. Also, according to the Saha ionization equilibrium, the ratio $\frac{n_{X\ II} n_e}{n_{X\ I}}$ is a function of T. The analysis of $e_r(x_i)$ as a function of T leads to the same conclusion as for Fig. 13: the value of the temperature has very little impact on the diagnosis. We therefore conclude that the thermodynamic conditions (n_e, T) do not affect the predictions, whatever the species i and the spectral interval considered.

4.3. Execution time

The long-term objective being to carry out field diagnostics, this type of approach must be possible on machines with low

computing capacity. As such, the time measurements presented in this section are obtained from a standard laptop for IDLE CPU executions under Ubuntu 22.04.3LTS (Python 3.7.3 - all libraries up to date on Aug. 2024).

In order to best simulate field conditions, we consider 4 distinct recurring tasks in the CF-LIBS quantification procedure.

1. Writing and reading an image (1024x1024) in proprietary .spe format (acquired from the Lightfield software). This image (X,Y) also contains all the metadata linked to the experiment carried out.
2. Calibration in spectral radiance and wavelength from this metadata in order to be able to establish a physical comparison between the observed and modeled emission. We obtain an image (λ, Y) whose pixel value is expressed in radiance L_λ.
3. Spatial integration on a portion of the sensor so as to obtain the experimentally studied spectrum $L_\lambda = f(\lambda)$.
4. Submission of this spectrum to the network in order to perform quantification. This last step includes:
    - normalization of the spectrum according to the Z-score normalization of Ω (see section 2.1)
    - passage in the network
    - converting network outputs to physical values

Process	Mean duration (s)	Deviation (s)
1	0.0070	0.0027
2	0.0292	0.0033
3	0.0029	0.0002
4	0.0453	0.0013
$1, 2, 3, 4$	**0.0844**	**0.0075**

Table 3: Statistics over 1000 IDLE CPU runs for representative experimental protocol on (i7-8750H@2.20GHz, 32Go-DDR4@3.2GHz, M2.2280-NVMe@32Gbit/s) machine.

We repeat these four operations 1000 times in order to extract the characteristic times exposed in Table 3. The total duration of the procedure (tasks 1, 2, 3 and 4) is measured at (0.0844 ± 0.0075) s. This demonstrates that in the current state of development, the optimization allows a CF-LIBS quantification rate of 10 Hz (repetition rate of most Nd-YAG lasers used for LIBS). This means that in GPU execution, we can reasonably expect to maintain a frame rate > 100 Hz. At the current development pace, high-speed applications (kHz) such as μ-LIBS will be possible in the coming months and we will thus be able to offer elementary mapping in CF-LIBS absolute quantification.

5. Conclusion

The work reported in this paper demonstrates the feasibility of an absolute quantification diagnosis by CF-LIBS in real time (10 Hz) using a machine learning approach. Despite the heavy computational capacities required to train such a generalized model, this type of approach particularly well suited for the chemical analysis of complex mixture still needs development regarding trace element among multiple majors. We do not explain here the choice of hyper parameters used to train our network. Indeed this is an exploratory work, and we are currently tuning those hyper parameters for better results. We show that the developed network presents, depending on the electronic structure of the species considered, a strong sensitivity to the spectral interval considered. Conversely, the diagnosis does not appear very sensitive to the thermodynamic conditions of the spectrum studied, whatever the species considered. By performing favorable spectral filtering, we show that the simultaneous quantification of Ag, Ar, B, C, Cu, H, Mo, Ni and W can be done to within ~ 10% whatever the values of the electron density and the temperature. Although these results are promising, it should be noted that this is a test-case carried out using synthetic spectra. Indeed, we plan in the short term to carry out an experimental validation campaign on calibrated samples in order to evaluate 1) the influence of uncertainties related to spectroscopic constants, in particular the probabilities of spontaneous emission as well as the Stark broadening and shift parameters, and 2) the influence of acquisition fluctuations.

Appendix

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Declaration of interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Credit author statement

Aurélien Favre: investigation, development, writing, **Alexis Abad**: development, writing, **Alexandre Poux**: development, **Léo Gosse**: investigation, **Ahmad Berjaoui**: reviewing, **Vincent Morel**: supervision, **Arnaud Bultel**: ressources.

Table A.1: Roman variables used in the paper

Constant	Meaning	Unit
c_i	Prediction confidence of the variable i	
$\bar{c}(i)$	Mean confidence over N_k predictions of the variable i	
$c_P(i)$	Pearson coefficient between p_i and t_i	
$e_r(i)$	Absolute deviation of between p_i and t_i	
$\bar{e}_r(i)$	Mean absolute deviation between p_i and t_i (over N_k comparisons)	
i_{max}	Number of targeted output	
ℓ_c	Core dimension of emitting plasma	m
$\mathcal{L}$	Customized confidence-based loss function	
L_λ	Spectral radiance	W m^{-2} sr^{-1} m^{-1}
$L^0_{\lambda,T}$	Black body spectral radiance at temperature T	W m^{-2} sr^{-1} m^{-1}
n_e	Electron density	m^{-3}
p_i	Prediction of the variable i	
t_i	Target value of the variable i	
T	Temperature	K
x_i	Mole fraction of element i	
x_i^*	Mole fraction of element i in log ratio space	

Table A.2: Greek variables used in the paper

Variable	Meaning	Unit
α_λ	Spectral absorption coefficient	m^{-1}
λ_c	Central wavelength	m
σ_{Ω_L}	Standard deviation of Ω_L	
Ω	Complete raw from computation dataset	
Ω^*	Normalized dataset	
$\check{\Omega}^*$	Randomly selected 0.1% sample of Ω^*	
$\check{\Omega}^*_f$	Randomly $630 < \lambda_c(\text{nm}) < 670$ selected 0.1% sample of Ω^*	
Ω_{λ_c}	Fraction of Ω containing raw from computation λ_c values	
$\Omega^*_{\lambda_c}$	Z-score normalized Ω_{λ_c}	
Ω_L	Fraction of Ω containing $\log(\mathbf{L}_\lambda)$ vectors	
Ω^*_L	Z-score normalized Ω_L	
$\bar{\Omega}_L$	Arithmetic mean of Ω_L	
Ω_{L_λ}	Fraction of Ω containing raw from computation $\mathbf{L}_\lambda$ vectors	
Ω_{n_e}	Fraction of Ω containing raw from computation n_e values	
$\Omega^*_{n_e}$	Z-score normalized Ω_{n_e}	
Ω_T	Fraction of Ω containing raw from computation T values	
Ω^*_T	Z-score normalized Ω_T	
Ω_{x_i}	Fraction of Ω containing raw from computation $\mathbf{x_i}$ vectors	
$\Omega_{x_i^*}$	Fraction of Ω containing log ratio $\mathbf{x_i^*}$ vectors	
$\Omega^*_{x_i^*}$	Z-score normalized $\Omega_{x_i^*}$	

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